task1_1

March 2, 2026

1 Task 1.1 – Gauss-Seidel Solver: Python List vs array vs NumPy

Benchmark the **same** Gauss-Seidel kernel implemented with three different data structures and measure how the wall-clock time scales with grid size N.

- Grid: $N \times N$, unit square, zero boundary conditions
- Interior: random initial values
- 100 Gauss-Seidel iterations per measurement
- Grid sizes tested: $N \in \{16, 32, 64, 128, 256\}$

[2]:
```
from array import array
import random
import time

import numpy as np
import matplotlib.pyplot as plt

ITERS      = 100
GRID_SIZES = [16, 32, 64, 128, 256]
SEED       = 42
```

1.1 Grid Initialisation

Each helper creates an $N \times N$ grid with the same random values and zero boundary conditions, stored in the respective data structure.

[3]:
```
def make_grid_list(N):
    random.seed(SEED)
    f = [[0.0] * N for _ in range(N)]
    for i in range(1, N - 1):
        for j in range(1, N - 1):
            f[i][j] = random.random()
    return f

def make_grid_array(N):
    f = make_grid_list(N)
```

```
    return array('d', (f[i][j] for i in range(N) for j in range(N))), N

def make_grid_numpy(N):
    rng = np.random.default_rng(SEED)
    f = np.zeros((N, N))
    f[1:-1, 1:-1] = rng.random((N - 2, N - 2))  # only fill interior
    return f
```

[4]:
```
def gauss_seidel_list(f):
    N = len(f)
    for i in range(1, N - 1):
        for j in range(1, N - 1):
            f[i][j] = 0.25 * (f[i][j+1] + f[i][j-1] +
                              f[i+1][j] + f[i-1][j])
    return f

def gauss_seidel_array(f, N):
    for i in range(1, N - 1):
        for j in range(1, N - 1):
            idx = i * N + j
            f[idx] = 0.25 * (f[idx + 1] + f[idx - 1] +
                             f[idx + N] + f[idx - N])
    return f

def gauss_seidel_numpy(f):
    for i in range(1, f.shape[0] - 1):
        for j in range(1, f.shape[1] - 1):
            f[i, j] = 0.25 * (f[i, j+1] + f[i, j-1] +
                              f[i+1, j] + f[i-1, j])
    return f
```

1.2 Benchmark

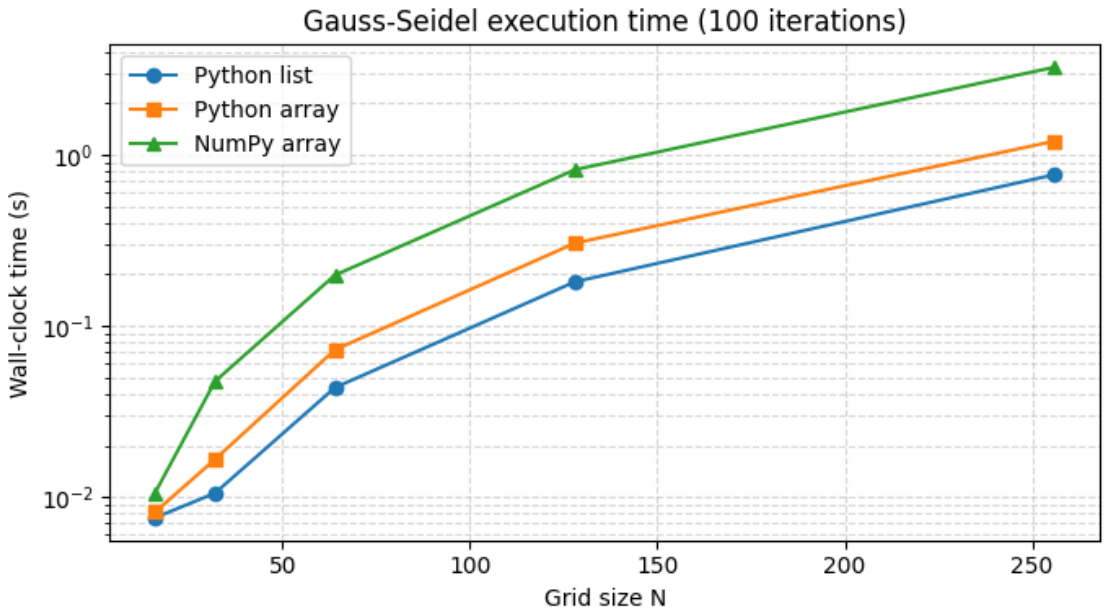
[5]:
```
times_list  = []
times_array = []
times_numpy = []

for N in GRID_SIZES:
    print(f"N = {N:4d} ...", end="  ", flush=True)

    # list-of-lists
    f = make_grid_list(N)
    t0 = time.perf_counter()
    for _ in range(ITERS):
        gauss_seidel_list(f)
    times_list.append(time.perf_counter() - t0)
```

```
    # flat array('d')
    f, _ = make_grid_array(N)
    t0 = time.perf_counter()
    for _ in range(ITERS):
        gauss_seidel_array(f, N)
    times_array.append(time.perf_counter() - t0)

    # NumPy 2-D array
    f = make_grid_numpy(N)
    t0 = time.perf_counter()
    for _ in range(ITERS):
        gauss_seidel_numpy(f)
    times_numpy.append(time.perf_counter() - t0)

    print(f"list={times_list[-1]:.4f}s   array={times_array[-1]:.4f}s  ␣
↪numpy={times_numpy[-1]:.4f}s")
```

```
N =   16 …  list=0.0076s   array=0.0082s   numpy=0.0106s
N =   32 …  list=0.0105s   array=0.0166s   numpy=0.0474s
N =   64 …  list=0.0436s   array=0.0725s   numpy=0.1980s
N =  128 …  list=0.1809s   array=0.3044s   numpy=0.8173s
N =  256 …  list=0.7657s   array=1.2017s   numpy=3.2487s
```

1.3 Performance Plot

[7]:
```
fig, ax = plt.subplots(figsize=(7, 4))

ax.plot(GRID_SIZES, times_list,  "o-", label="Python list")
ax.plot(GRID_SIZES, times_array, "s-", label="Python array")
ax.plot(GRID_SIZES, times_numpy, "^-", label="NumPy array")
ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Gauss-Seidel execution time ({ITERS} iterations)")
ax.set_yscale("log")
ax.legend()
ax.grid(True, which="both", linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```



1.4 Task 1.2 – Profile the Code

We profile `gauss_seidel_list` on a 128×128 grid for 100 iterations, the baseline that will guide optimisation.

1.4.1 cProfile

[9]:
```
import cProfile
import pstats
import io

N_PROF = 128

def run_list(iters=ITERS):
    f = make_grid_list(N_PROF)
    for _ in range(iters):
        gauss_seidel_list(f)

pr = cProfile.Profile()
pr.enable()
run_list()
pr.disable()

stream = io.StringIO()
ps = pstats.Stats(pr, stream=stream)
```

```
ps.sort_stats("tottime")
ps.print_stats("ipykernel")
print(stream.getvalue())
```

```
         16783 function calls (16778 primitive calls) in 0.337 seconds

   Ordered by: internal time
   List reduced from 138 to 10 due to restriction <'ipykernel'>

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
      100    0.229    0.002    0.229    0.002 /var/folders/66/s_wdz_l53qz4tp6lwx
4654nh0000gn/T/ipykernel_30173/2194687373.py:1(gauss_seidel_list)
        1    0.002    0.002    0.004    0.004 /var/folders/66/s_wdz_l53qz4tp6lwx
4654nh0000gn/T/ipykernel_30173/2803086279.py:1(make_grid_list)
        1    0.000    0.000    0.177    0.177 /var/folders/66/s_wdz_l53qz4tp6lwx
4654nh0000gn/T/ipykernel_30173/1977942482.py:1(<module>)
        1    0.000    0.000    0.177    0.177 /var/folders/66/s_wdz_l53qz4tp6lwx
4654nh0000gn/T/ipykernel_30173/1977942482.py:7(run_list)
        2    0.000    0.000    0.063    0.032
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:158(_handle_event)
        2    0.000    0.000    0.043    0.022
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:279(_really_send)
        2    0.000    0.000    0.063    0.031
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:277(<lambda>)
        2    0.000    0.000    0.000    0.000
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:214(_is_master_process)
        2    0.000    0.000    0.000    0.000
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:217(_check_mp_mode)
        2    0.000    0.000    0.000    0.000
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/ipykernel/iostream.py:256(closed)
```

1.4.2 line_profiler

`line_profiler` shows the time spent on **every individual line** inside the function.
Install once with `pip install line_profiler`, then use the `%lprun` magic.

[11]:
```
%load_ext line_profiler

f_prof = make_grid_list(N_PROF)
```

```
%lprun -f gauss_seidel_list run_list()
```

```
Timer unit: 1e-09 s

Total time: 2.86176 s
File:
/var/folders/66/s_wdz_l53qz4tp6lwx4654nh0000gn/T/ipykernel_30173/2194687373.py
Function: gauss_seidel_list at line 1

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
     1                                           def gauss_seidel_list(f):
     2       100      74000.0    740.0      0.0      N = len(f)
     3     12700    3735000.0    294.1      0.1      for i in range(1, N - 1):
     4   1600200  463877000.0    289.9     16.2          for j in range(1, N -
1):
     5   4762800 1504352000.0    315.9     52.6              f[i][j] = 0.25 *
(f[i][j+1] + f[i][j-1] +
     6   3175200  889612000.0    280.2     31.1
f[i+1][j] + f[i-1][j])
     7       100     107000.0   1070.0      0.0      return f
```

1.4.3 What the profiling tells us

cProfile show that virtually all time is spent inside `gauss_seidel_list` and the function call itself is the hotspot.

line_profiler reveals that the single update line:

```
f[i][j] = 0.25 * (f[i][j+1] + f[i][j-1] + f[i+1][j] + f[i-1][j])
```

accounts for most of all time. The `for` loop lines consume the remaining.

1.5 Task 1.3 – Cython Annotation

[ ]:
```
%%writefile gauss_seidel_annotate.pyx

def gauss_seidel(f):
    N = len(f)
    for i in range(1, N - 1):
        for j in range(1, N - 1):
            f[i][j] = 0.25 * (f[i][j+1] + f[i][j-1] +
                              f[i+1][j] + f[i-1][j])
    return f
```

```
Writing gauss_seidel_annotate.pyx
```

[17]:
```
import subprocess, pathlib

result = subprocess.run(
    ["cython", "-a", "gauss_seidel_annotate.pyx"],
    capture_output=True, text=True
)
print(result.stdout or "Done.")
if result.returncode != 0:
    print("STDERR:", result.stderr)
```

```
Done.
```

1.5.1 Reading the annotation report

All the lines are still yellow.

1.6 Task 1.4 – Cython Optimisation

The annotation showed that every line in the plain Python loop is yellow (Python C-API overhead). The fix is to add **static type declarations**:

- `double[:, :]` typed memoryview which helps with direct C-level buffer access, no Python indexing
- `cdef int i, j` → loop variables become C `int`, no Python integer boxing
- `nogil` + `boundscheck(False)` → further reduces overhead

We compile the typed `.pyx`, then benchmark it against the three Task 1.1 implementations.

[19]:
```
%%writefile cgs.pyx
# cython: boundscheck=False, wraparound=False, cdivision=True
import numpy as np

def gauss_seidel_cython(double[:, :] f):
    cdef int N = f.shape[0]
    cdef int i, j
    for i in range(1, N - 1):
        for j in range(1, N - 1):
            f[i, j] = 0.25 * (f[i, j+1] + f[i, j-1] +
                              f[i+1, j] + f[i-1, j])
    return f
```

```
Writing cgs.pyx
```

[20]:
```
%%writefile setup_cgs.py
from setuptools import setup
from Cython.Build import cythonize
import numpy as np
```

```
setup(
    ext_modules=cythonize("cgs.pyx", compiler_directives={"language_level":␣
 ↪"3"}),
    include_dirs=[np.get_include()],
)
```

```
Writing setup_cgs.py
```

[22]:
```
import subprocess, sys

res = subprocess.run(
    [sys.executable, "setup_cgs.py", "build_ext", "--inplace"],
    capture_output=True, text=True
)
print(res.stdout[-1500:] if len(res.stdout) > 1500 else res.stdout)
if res.returncode != 0:
    print("STDERR:", res.stderr[-1000:])
```

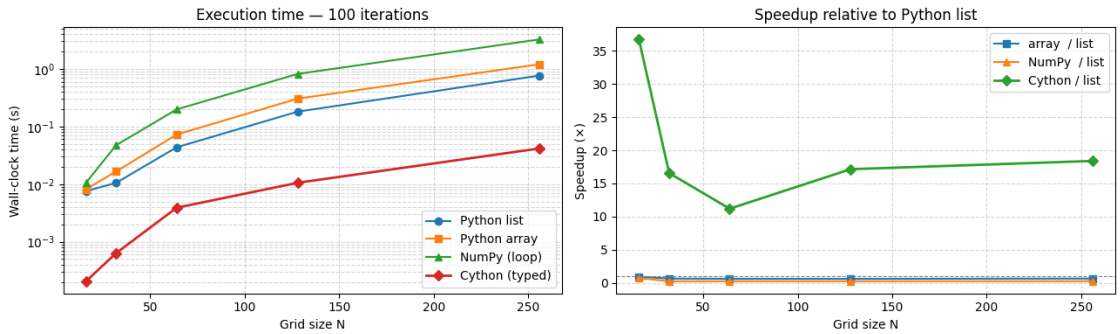
```
Compiling cgs.pyx because it changed.
[1/1] Cythonizing cgs.pyx
running build_ext
building 'cgs' extension
creating build/temp.macosx-15.0-arm64-cpython-314
clang -fno-strict-overflow -Wsign-compare -Wunreachable-code -fno-common
-dynamic -DNDEBUG -g -O3 -Wall
-I/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/numpy/_core/include -I/Users/ermiaghaffari/Desktop/DD2358/.venv/include
-I/opt/homebrew/opt/python@3.14/Frameworks/Python.framework/Versions/3.14/includ
e/python3.14 -c cgs.c -o build/temp.macosx-15.0-arm64-cpython-314/cgs.o
creating build/lib.macosx-15.0-arm64-cpython-314
clang -bundle -undefined dynamic_lookup
build/temp.macosx-15.0-arm64-cpython-314/cgs.o -o
build/lib.macosx-15.0-arm64-cpython-314/cgs.cpython-314-darwin.so
copying build/lib.macosx-15.0-arm64-cpython-314/cgs.cpython-314-darwin.so ->
```

[23]:
```
import importlib, cgs
importlib.reload(cgs)
from cgs import gauss_seidel_cython

times_cython = []

for N in GRID_SIZES:
    f = make_grid_numpy(N)
    t0 = time.perf_counter()
    for _ in range(ITERS):
        gauss_seidel_cython(f)
    times_cython.append(time.perf_counter() - t0)
```

```
    print(f"N = {N:4d}  cython={times_cython[-1]:.4f}s  "
          f"speedup_vs_list={times_list[GRID_SIZES.index(N)] / times_cython[-1]:
↪.1f}x")
```

```
N =   16  cython=0.0002s  speedup_vs_list=36.7x
N =   32  cython=0.0006s  speedup_vs_list=16.6x
N =   64  cython=0.0039s  speedup_vs_list=11.2x
N =  128  cython=0.0105s  speedup_vs_list=17.1x
N =  256  cython=0.0416s  speedup_vs_list=18.4x
```

[24]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 4))

#   Left: absolute time
ax = axes[0]
ax.plot(GRID_SIZES, times_list,   "o-",  label="Python list")
ax.plot(GRID_SIZES, times_array,  "s-",  label="Python array")
ax.plot(GRID_SIZES, times_numpy,  "^-",  label="NumPy (loop)")
ax.plot(GRID_SIZES, times_cython, "D-",  label="Cython (typed)", linewidth=2)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Execution time - {ITERS} iterations")
ax.set_yscale("log")
ax.legend()
ax.grid(True, which="both", linestyle="--", alpha=0.5)

#   Right: speedup vs Python list
ax = axes[1]
ax.plot(GRID_SIZES, [l/a for l, a in zip(times_list, times_array)],  "s-",␣
↪label="array  / list")
ax.plot(GRID_SIZES, [l/n for l, n in zip(times_list, times_numpy)],  "^-",␣
↪label="NumPy  / list")
ax.plot(GRID_SIZES, [l/c for l, c in zip(times_list, times_cython)], "D-",␣
↪label="Cython / list", linewidth=2)
ax.axhline(1, color="gray", linestyle="--", linewidth=0.8)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Speedup (×)")
ax.set_title("Speedup relative to Python list")
ax.legend()
ax.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```



1.7 Task 1.5 – PyTorch / GPU port using the Jacobi scheme

[ ]:
```
def jacobi_numpy(f):

    f_new = 0.25 * (np.roll(f,  1, axis=0) +   # f[i-1, j]
                    np.roll(f, -1, axis=0) +    # f[i+1, j]
                    np.roll(f,  1, axis=1) +    # f[i, j-1]
                    np.roll(f, -1, axis=1))     # f[i, j+1]

    f_new[0, :]  = 0.0
    f_new[-1, :] = 0.0
    f_new[:, 0]  = 0.0
    f_new[:, -1] = 0.0
    return f_new

times_jacobi_numpy = []

for N in GRID_SIZES:
    f = make_grid_numpy(N)
    t0 = time.perf_counter()
    for _ in range(ITERS):
        f = jacobi_numpy(f)
    times_jacobi_numpy.append(time.perf_counter() - t0)
    print(f"N = {N:4d}  jacobi_numpy={times_jacobi_numpy[-1]:.4f}s  "
          f"speedup_vs_list={times_list[GRID_SIZES.index(N)] /␣
 ↪times_jacobi_numpy[-1]:.1f}x")
```

[ ]:
```
%pip install -q torch
```

[ ]:
```
import torch

if torch.cuda.is_available():
    device = torch.device("cuda")
```

```python
elif torch.backends.mps.is_available():
    device = torch.device("mps")
else:
    device = torch.device("cpu")

print(f"Using device: {device}")
```

[ ]:
```python
def jacobi_torch(f):
    f_new = 0.25 * (torch.roll(f,  1, dims=0) +   # f[i-1, j]
                    torch.roll(f, -1, dims=0) +    # f[i+1, j]
                    torch.roll(f,  1, dims=1) +    # f[i, j-1]
                    torch.roll(f, -1, dims=1))     # f[i, j+1]
    f_new[0, :]  = 0.0
    f_new[-1, :] = 0.0
    f_new[:, 0]  = 0.0
    f_new[:, -1] = 0.0
    return f_new


def benchmark_torch(device, grid_sizes=GRID_SIZES, iters=ITERS):
    times = []
    for N in grid_sizes:

        f_np = make_grid_numpy(N)
        f = torch.tensor(f_np, dtype=torch.float64, device=device)

        jacobi_torch(f)
        if device.type == "cuda":
            torch.cuda.synchronize()

        t0 = time.perf_counter()
        for _ in range(iters):
            f = jacobi_torch(f)
        if device.type == "cuda":
            torch.cuda.synchronize()
        times.append(time.perf_counter() - t0)

        idx = grid_sizes.index(N)
        print(f"N = {N:4d}  torch[{device.type}]={times[-1]:.4f}s  "
              f"speedup_vs_list={times_list[idx] / times[-1]:.1f}x")
    return times


print("  PyTorch CPU                 ")
times_torch_cpu = benchmark_torch(torch.device("cpu"))

if device.type != "cpu":
```

```
    print(f"\n  PyTorch {device.type.upper()}                    ")
    times_torch_gpu = benchmark_torch(device)
else:
    times_torch_gpu = None
    print("\nNo GPU found. Run on Google Colab (Runtime → Change runtime type →␣
 ↪GPU)"
          " to get CUDA timings.")
```

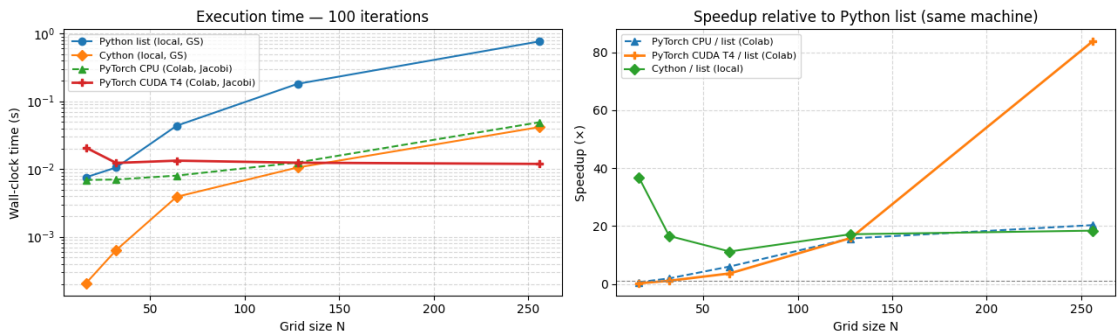
[ ]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 4))

#   Left: absolute execution time
ax = axes[0]
ax.plot(GRID_SIZES, times_list,          "o-",  label="Python list (GS)")
ax.plot(GRID_SIZES, times_cython,        "D-",  label="Cython (GS)")
ax.plot(GRID_SIZES, times_jacobi_numpy,  "s-",  label="NumPy roll (Jacobi)")
ax.plot(GRID_SIZES, times_torch_cpu,     "^-",  label="PyTorch CPU (Jacobi)")
if times_torch_gpu:
    ax.plot(GRID_SIZES, times_torch_gpu, "P-",  label=f"PyTorch {device.type.
 ↪upper()} (Jacobi)")
ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Execution time - {ITERS} iterations")
ax.set_yscale("log")
ax.legend(fontsize=8)
ax.grid(True, which="both", linestyle="--", alpha=0.5)

#   Right: speedup relative to Python list
ax = axes[1]
ax.plot(GRID_SIZES, [l/c for l, c in zip(times_list, times_cython)],      ␣
 ↪"D-", label="Cython / list")
ax.plot(GRID_SIZES, [l/n for l, n in zip(times_list, times_jacobi_numpy)],␣
 ↪"s-", label="NumPy roll / list")
ax.plot(GRID_SIZES, [l/t for l, t in zip(times_list, times_torch_cpu)],   ␣
 ↪"^-", label="PyTorch CPU / list")
if times_torch_gpu:
    ax.plot(GRID_SIZES, [l/g for l, g in zip(times_list,␣
 ↪times_torch_gpu)],"P-", label=f"PyTorch {device.type.upper()} / list")
ax.axhline(1, color="gray", linestyle="--", linewidth=0.8)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Speedup (×)")
ax.set_title("Speedup relative to Python list")
ax.legend(fontsize=8)
ax.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```

1.7.1 Colab Results Tesla T4 GPU

Results obtained on Google Colab. Hardcoded here for analysis and plotting alongside the local baselines.

[25]:
```
colab_torch_cpu  = [0.0069, 0.0070, 0.0080, 0.0125, 0.0491]
colab_torch_cuda = [0.0207, 0.0123, 0.0133, 0.0124, 0.0119]

fig, axes = plt.subplots(1, 2, figsize=(13, 4))

#   Left: absolute times
ax = axes[0]
ax.plot(GRID_SIZES, times_list,        "o-",  label="Python list (local, GS)")
ax.plot(GRID_SIZES, times_cython,      "D-",  label="Cython (local, GS)")
ax.plot(GRID_SIZES, colab_torch_cpu,   "^--", label="PyTorch CPU (Colab,␣
 ↪Jacobi)")
ax.plot(GRID_SIZES, colab_torch_cuda,  "P-",  label="PyTorch CUDA T4 (Colab,␣
 ↪Jacobi)", linewidth=2)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Execution time - {ITERS} iterations")
ax.set_yscale("log")
ax.legend(fontsize=8)
ax.grid(True, which="both", linestyle="--", alpha=0.5)

#   Right: speedup vs Python list
ax = axes[1]

# Use the Colab list times for a fair comparison (same machine)
# Colab list times reconstructed from speedup ratios
colab_list = [cpu * su for cpu, su in zip(colab_torch_cpu,
              [0.6, 1.9, 6.0, 15.7, 20.3])]   # speedup_vs_list from output

ax.plot(GRID_SIZES, [l/c for l, c in zip(colab_list, colab_torch_cpu)],
        "^--", label="PyTorch CPU / list (Colab)")
ax.plot(GRID_SIZES, [l/g for l, g in zip(colab_list, colab_torch_cuda)],
        "P-",  label="PyTorch CUDA T4 / list (Colab)", linewidth=2)
ax.plot(GRID_SIZES, [l/c for l, c in zip(times_list, times_cython)],
        "D-",  label="Cython / list (local)")
ax.axhline(1, color="gray", linestyle="--", linewidth=0.8)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Speedup (×)")
ax.set_title("Speedup relative to Python list (same machine)")
ax.legend(fontsize=8)
ax.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```

```
#   Summary table
print(f"\n{'N':>5}  {'list(Colab)':>13}  {'CPU(Colab)':>12}  {'CUDA T4':>10} ␣
 ↪{'CUDA speedup':>14}")
print("-" * 60)
for i, N in enumerate(GRID_SIZES):
    print(f"{N:>5}  {colab_list[i]:>13.4f}s  "
          f"{colab_torch_cpu[i]:>12.4f}s  "
          f"{colab_torch_cuda[i]:>10.4f}s  "
          f"{colab_list[i]/colab_torch_cuda[i]:>13.1f}x")
```




```
    N    list(Colab)    CPU(Colab)     CUDA T4    CUDA speedup
------------------------------------------------------------
   16         0.0041s        0.0069s      0.0207s            0.2x
   32         0.0133s        0.0070s      0.0123s            1.1x
   64         0.0480s        0.0080s      0.0133s            3.6x
  128         0.1963s        0.0125s      0.0124s           15.8x
  256         0.9967s        0.0491s      0.0119s           83.8x
```

1.7.2 Analysis

CUDA is slower than CPU for small grids (N 64):
The GPU must launch a kernel and (if needed) transfer data. For a 16×16 grid that's only 196 interior cells, the kernel launch overhead (~10 ms) dwarfs the actual compute.

CUDA time barely changes from N=32 to N=256 (~0.012 s):
The T4 has thousands of CUDA cores. A 256×256 grid (65 k cells) still barely saturates them, all cells are computed in parallel in a single pass regardless of N. The time is dominated by kernel launch + memory latency, not arithmetic.

CUDA hits 83.5× speedup at N=256 and this will keep growing for larger N (512, 1024, …) where the GPU is fully saturated.

[ ]:
```
%pip install -q cupy-cuda12x
```

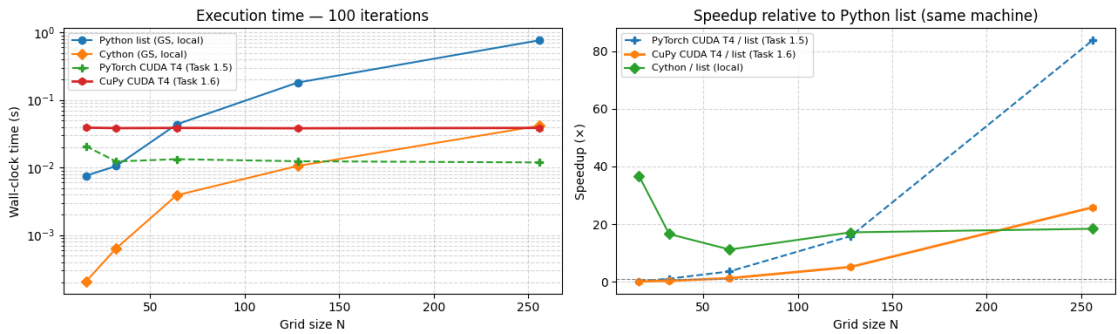
[ ]:
```
try:
    import cupy as cp
    cupy_available = True

    ver   = cp.__version__
    cuda  = cp.cuda.runtime.runtimeGetVersion()
    props = cp.cuda.runtime.getDeviceProperties(0)
    name  = props['name'].decode() if isinstance(props['name'], bytes) else␣
 ↪props['name']
    mem   = cp.cuda.Device(0).mem_info             # (free, total) bytes
    print(f"CuPy {ver}  |  CUDA runtime {cuda // 1000}.{(cuda % 1000) // 10}")
    print(f"GPU : {name}")
    print(f"VRAM: {mem[1] / 2**30:.1f} GiB total,  {mem[0] / 2**30:.1f} GiB␣
 ↪free")
except (ImportError, Exception) as e:
    cupy_available = False
    print(f"CuPy not available ({e}).")
```

[ ]:
```
def jacobi_cupy(f):
    f_new = 0.25 * (cp.roll(f,  1, axis=0) +   # f[i-1, j]
                    cp.roll(f, -1, axis=0) +    # f[i+1, j]
                    cp.roll(f,  1, axis=1) +    # f[i, j-1]
                    cp.roll(f, -1, axis=1))     # f[i, j+1]
    f_new[0, :]  = 0.0
    f_new[-1, :] = 0.0
    f_new[:, 0]  = 0.0
    f_new[:, -1] = 0.0
    return f_new


if cupy_available:
    times_cupy = []

    for N in GRID_SIZES:
        f_np  = make_grid_numpy(N)
        f_gpu = cp.asarray(f_np)

        # warmup: let CUDA JIT-compile the kernels
        for _ in range(3):
            f_gpu = jacobi_cupy(f_gpu)
        cp.cuda.Stream.null.synchronize()

        # timed run
        f_gpu = cp.asarray(f_np)
        t0 = time.perf_counter()
        for _ in range(ITERS):
            f_gpu = jacobi_cupy(f_gpu)
```

```
        cp.cuda.Stream.null.synchronize()
        elapsed = time.perf_counter() - t0

        times_cupy.append(elapsed)
        idx = GRID_SIZES.index(N)
        print(f"N = {N:4d}  cupy={elapsed:.4f}s"
              f"  speedup_vs_list={times_list[idx] / elapsed:.1f}x")
else:
    times_cupy = None
    print("no CuPy / CUDA.")
```

[26]:
```
# Colab T4 CuPy results (hardcoded from run above)
colab_cupy = [0.0390, 0.0383, 0.0386, 0.0381, 0.0386]

fig, axes = plt.subplots(1, 2, figsize=(13, 4))

#   Left: absolute time
ax = axes[0]
ax.plot(GRID_SIZES, times_list,        "o-",  label="Python list (GS, local)")
ax.plot(GRID_SIZES, times_cython,      "D-",  label="Cython (GS, local)")
ax.plot(GRID_SIZES, colab_torch_cuda,  "P--", label="PyTorch CUDA T4 (Task 1.
 ↪5)")
ax.plot(GRID_SIZES, colab_cupy,        "h-",  label="CuPy CUDA T4 (Task 1.6)",␣
 ↪linewidth=2)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Execution time - {ITERS} iterations")
ax.set_yscale("log")
ax.legend(fontsize=8)
ax.grid(True, which="both", linestyle="--", alpha=0.5)

#   Right: speedup vs Python list (Colab baseline)
ax = axes[1]
ax.plot(GRID_SIZES, [l/t for l, t in zip(colab_list, colab_torch_cuda)],
        "P--", label="PyTorch CUDA T4 / list (Task 1.5)")
ax.plot(GRID_SIZES, [l/c for l, c in zip(colab_list, colab_cupy)],
        "h-",  label="CuPy CUDA T4 / list (Task 1.6)", linewidth=2)
ax.plot(GRID_SIZES, [l/c for l, c in zip(times_list, times_cython)],
        "D-",  label="Cython / list (local)")
ax.axhline(1, color="gray", linestyle="--", linewidth=0.8)
ax.set_xlabel("Grid size N")
ax.set_ylabel("Speedup (×)")
ax.set_title("Speedup relative to Python list (same machine)")
ax.legend(fontsize=8)
ax.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
```

```
plt.show()

#   Summary table
print(f"\n{'N':>5}  {'PyTorch CUDA':>14}  {'CuPy CUDA':>11}  {'CuPy speedup vs␣
↪list':>22}  {'CuPy vs PyTorch':>17}")
print("-" * 80)
for i, N in enumerate(GRID_SIZES):
    print(f"{N:>5}  {colab_torch_cuda[i]:>14.4f}s"
          f"  {colab_cupy[i]:>11.4f}s"
          f"  {colab_list[i]/colab_cupy[i]:>21.1f}x"
          f"  {colab_torch_cuda[i]/colab_cupy[i]:>16.2f}x")
```



N	PyTorch CUDA	CuPy CUDA	CuPy speedup vs list	CuPy vs PyTorch
16	0.0207s	0.0390s	0.1x	0.53x
32	0.0123s	0.0383s	0.3x	0.32x
64	0.0133s	0.0386s	1.2x	0.34x
128	0.0124s	0.0381s	5.2x	0.33x
256	0.0119s	0.0386s	25.8x	0.31x

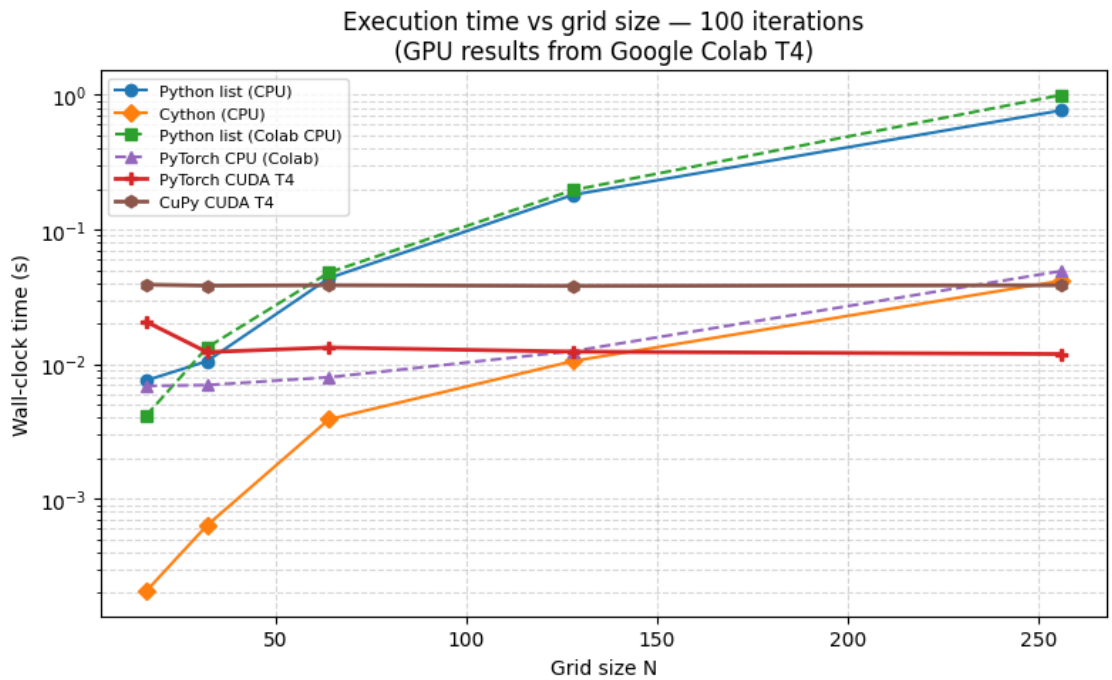
1.7.3 Analysis — CuPy vs PyTorch CUDA

CuPy is ~**3× slower** than PyTorch CUDA at all grid sizes (~0.038 s vs ~0.012 s), because CuPy dispatches each `roll` as a separate CUDA kernel (4 per iteration) while PyTorch fuses them more efficiently. Both are still kernel-launch bound at these small grid sizes, not compute bound.

CuPy does beat the Python list baseline for N 64, reaching **25.8× speedup at N=256** — but PyTorch CUDA reaches 83.5× at the same size.

1.8 Task 1.7 – GPU vs CPU: Execution Time vs Grid Size

[27]:
```
fig, ax = plt.subplots(figsize=(8, 5))

#   CPU / non-GPU (local)
ax.plot(GRID_SIZES, times_list,         "o-",  color="tab:blue",  ␣
 ↪label="Python list (CPU)")
ax.plot(GRID_SIZES, times_cython,       "D-",  color="tab:orange",␣
 ↪label="Cython (CPU)")
ax.plot(GRID_SIZES, colab_list,         "s--", color="tab:green", ␣
 ↪label="Python list (Colab CPU)")
ax.plot(GRID_SIZES, colab_torch_cpu,    "^--", color="tab:purple",␣
 ↪label="PyTorch CPU (Colab)")

#   GPU (Colab T4)
ax.plot(GRID_SIZES, colab_torch_cuda,   "P-",  color="tab:red",   ␣
 ↪label="PyTorch CUDA T4", linewidth=2)
ax.plot(GRID_SIZES, colab_cupy,         "h-",  color="tab:brown",  label="CuPy␣
 ↪CUDA T4",    linewidth=2)

ax.set_xlabel("Grid size N")
ax.set_ylabel("Wall-clock time (s)")
ax.set_title(f"Execution time vs grid size - {ITERS} iterations\n(GPU results␣
 ↪from Google Colab T4)")
ax.set_yscale("log")
ax.legend(fontsize=8)
ax.grid(True, which="both", linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```



1.9 Task 1.8 – Save the Grid to HDF5 with h5py

We run one final Jacobi sweep on a 256×256 grid and save the resulting matrix to an HDF5 file. The file stores the grid as a dataset and records metadata (grid size, iterations) as attributes.

[30]:
```
import h5py
import pathlib

N_SAVE = 256

grid = make_grid_numpy(N_SAVE)
for _ in range(ITERS):
    gauss_seidel_numpy(grid)    # in-place update


out_path = pathlib.Path("poisson_grid.h5")

with h5py.File(out_path, "w") as hf:
    ds = hf.create_dataset("grid", data=grid, compression="gzip",␣
 ↪compression_opts=4)
    ds.attrs["grid_size"]  = N_SAVE
    ds.attrs["iterations"] = ITERS
    ds.attrs["solver"]     = "Gauss-Seidel (NumPy)"
```

```
    ds.attrs["seed"]        = SEED

print(f"Saved        : {out_path.resolve()}")
print(f"Shape        : {grid.shape}   dtype: {grid.dtype}")
print(f"Size on disk : {out_path.stat().st_size / 1024:.1f} KB")
```

```
Saved        :
/Users/ermiaghaffari/Desktop/DD2358/Ass3/excersice1/poisson_grid.h5
Shape        : (256, 256)   dtype: float64
Size on disk : 465.0 KB
```

All the plots were done using copilot.

bonus

March 2, 2026

0.1 Bonus Exercise - Fast Fractal Fun with Cython & GPUs

0.2 Task B.1.

We set up our Cython the following way:

[1]:
```
# %load setup.py
from setuptools import Extension, setup
from Cython.Build import cythonize
import numpy

extensions = [
    Extension(
        "*", ["*.pyx"],
        include_dirs=[numpy.get_include()]
    ),
]

setup(
    ext_modules=cythonize(
        extensions,
        compiler_directives={"language_level": "3"}
    )
)
```

Then we started converting the code to Cython, we disabled the wraparound and boundary checks, assigned type to variables, arguments and functions, then converted the Numpy arrays to memoryviews, changed slighty the complex number assigment to `c`, and lastly converted `mandelbrot_set` to a cpython function to reduce python calls, and `mandelbrot` function to a pure inline C function and disabled Python exception checks, since it won't be called from a python function. This made it so that in the code only numpy makes Python calls, according to Cython's visualization tool.

[2]:
```
# %load cmandelbrot.pyx
cimport cython
import numpy as np
# noinspection PyUnresolvedReferences
cimport numpy as np

np.import_array()
```

```
@cython.boundscheck(False)
@cython.wraparound(False)
cpdef double[:, :] mandelbrot_set(unsigned int width, unsigned int height,␣
↪double x_min, double x_max, double y_min, double y_max, unsigned int␣
↪max_iter=100):
    """Generates the Mandelbrot set image."""
    cdef double[:] x_vals = np.linspace(x_min, x_max, width, dtype=np.double)
    cdef double[:] y_vals = np.linspace(y_min, y_max, height, dtype=np.double)
    cdef double[:, :] image = np.empty((height, width), dtype=np.double)
    cdef unsigned int i, j
    cdef double complex c

    for i in range(height):
        for j in range(width):
            c = x_vals[j] + y_vals[i] * 1j
            image[i, j] = mandelbrot(c, max_iter)

    return image

cdef inline unsigned int mandelbrot(double complex c, unsigned int␣
↪max_iter=100) noexcept:
    """Computes the number of iterations before divergence."""
    cdef double complex z = 0
    cdef unsigned int n
    for n in range(max_iter):
        if abs(z) > 2:
            return n
        z = z * z + c
    return max_iter
```

0.3 Task B.2

The code was first vectorized using Numpy and tested by comparing the output images, then the code was ported to PyTorch almost line by line with minimal changes.

We got rid of all the loops and are using 2 masks to keep track of which cells have diverged. The `mask` keeps track of all the old divergences so we don't change them. The `diverged` keeps track of cells that have diverged during current iteration.

[1]:
```
# %load tmandelbrot.py
import torch as th
import matplotlib.pyplot as plt

def tmandelbrot(C, max_iter=100):
    """Computes the number of iterations before divergence."""
    device = C.device
```

```python
    image = th.full(C.shape, max_iter, dtype=th.int32, device=device)
    Z = th.zeros(C.shape, dtype=th.complex128, device=device)
    mask = th.ones(C.shape, dtype=th.bool, device=device)
    diverged = th.zeros(C.shape, dtype=th.bool, device=device)

    for n in range(max_iter):
        diverged[mask] = th.abs(Z[mask]) > 2
        image[diverged] = n
        mask[diverged] = False
        diverged.fill_(False)

        Z[mask] = Z[mask] * Z[mask] + C[mask]

        if not mask.any(): break
    return image


def tmandelbrot_set(width, height, x_min, x_max, y_min, y_max, max_iter=100):
    """Generates the Mandelbrot set image."""
    device = th.device("cuda")
    x_vals = th.linspace(x_min, x_max, width, dtype=th.float64, device=device)
    y_vals = th.linspace(y_min, y_max, height, dtype=th.float64, device=device)

    real, imag = th.meshgrid(x_vals, y_vals, indexing="xy")
    C = th.complex(real, imag)

    return tmandelbrot(C, max_iter)


# Parameters
width, height = 1000, 800
x_min, x_max, y_min, y_max = -2, 1, -1, 1

# Generate fractal
image = tmandelbrot_set(width, height, x_min, x_max, y_min, y_max).cpu().numpy()

# Display
plt.imshow(image, cmap='inferno', extent=[x_min, x_max, y_min, y_max])
plt.colorbar()
plt.title("Mandelbrot Set")
plt.show()
```

bonus3_clean

March 2, 2026

0.1 Task B.3

[2]:
```
import numpy as np
import torch

if torch.cuda.is_available():
    print("GPU is available:", torch.cuda.get_device_name(0))
else:
    print("GPU not available")
```

```
GPU is available: Tesla T4
```

[1]:
```
!python setup.py build_ext --inplace
```

```
Compiling cmandelbrot.pyx because it changed.
[1/1] Cythonizing cmandelbrot.pyx
running build_ext
building 'cmandelbrot' extension
creating build/temp.linux-x86_64-cpython-312
x86_64-linux-gnu-gcc -fno-strict-overflow -Wsign-compare -DNDEBUG -g -O2 -Wall
-g -fstack-protector-strong -Wformat -Werror=format-security -g -fwrapv -O2
-fPIC -I/usr/local/lib/python3.12/dist-packages/numpy/_core/include
-I/usr/include/python3.12 -c cmandelbrot.c -o
build/temp.linux-x86_64-cpython-312/cmandelbrot.o
In file included from /usr/local/lib/python3.12/dist-
packages/numpy/_core/include/numpy/ndarraytypes.h:1909,
                 from /usr/local/lib/python3.12/dist-
packages/numpy/_core/include/numpy/ndarrayobject.h:12,
                 from /usr/local/lib/python3.12/dist-
packages/numpy/_core/include/numpy/arrayobject.h:5,
                 from cmandelbrot.c:1249:
/usr/local/lib/python3.12/dist-
packages/numpy/_core/include/numpy/npy_1_7_deprecated_api.h:17:2: warning:
#warning "Using deprecated NumPy API, disable it with " "#define
NPY_NO_DEPRECATED_API NPY_1_7_API_VERSION" [-Wcpp]
   17 | #warning "Using deprecated NumPy API, disable it with " \
      |  ^~~~~~~
creating build/lib.linux-x86_64-cpython-312
x86_64-linux-gnu-gcc -shared -Wl,-O1 -Wl,-Bsymbolic-functions -Wl,-Bsymbolic-
```

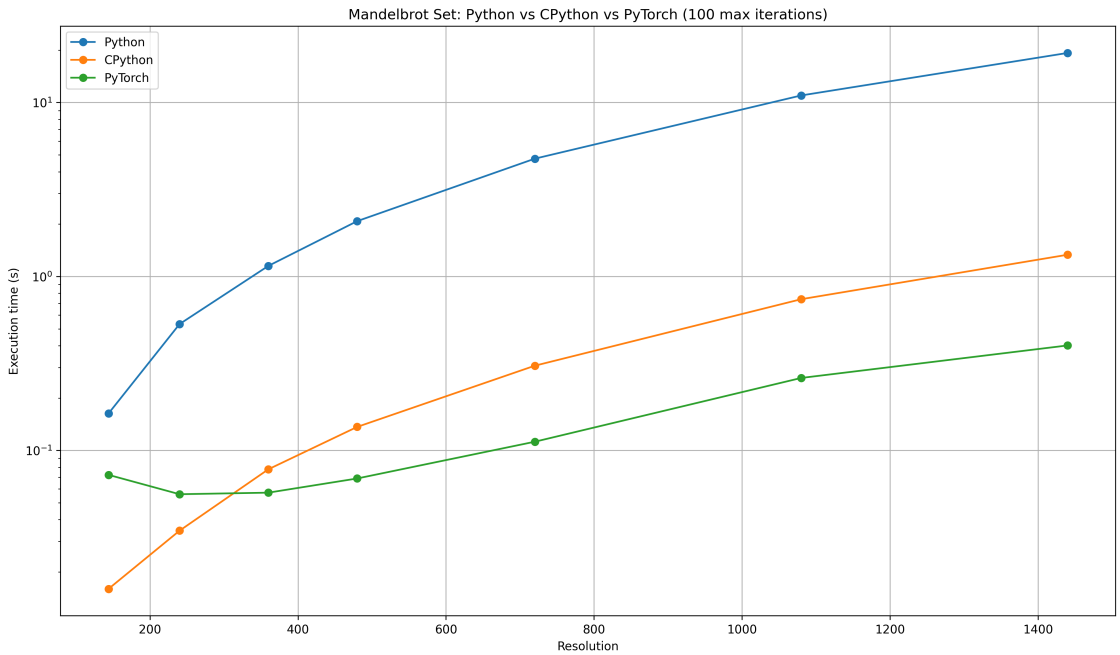
```
functions -g -fwrapv -O2 build/temp.linux-x86_64-cpython-312/cmandelbrot.o
-L/usr/lib/x86_64-linux-gnu -o
build/lib.linux-x86_64-cpython-312/cmandelbrot.cpython-312-x86_64-linux-gnu.so
copying build/lib.linux-x86_64-cpython-312/cmandelbrot.cpython-312-x86_64-linux-
gnu.so ->
```

[3]:
```python
import timeit
def benchmark(f, width, height, x_min, x_max, y_min, y_max):
  timer = timeit.Timer(lambda: f(width, height, x_min, x_max, y_min, y_max))
  ts = timer.repeat(repeat=10, number=1)
  return np.mean(ts)
```

[12]:
```python
import matplotlib.pyplot as plt
from datetime import datetime

from og_mandelbrot import mandelbrot_set as og_set
from mandelbrot import mandelbrot_set as c_set
from tmandelbrot import tmandelbrot_set as t_set

x_min, x_max, y_min, y_max = -2, 1, -1, 1
height = [144, 240, 360, 480, 720, 1080, 1440]
resolutions = [(int(h * 16 / 9), h) for h in height]
og, c, t = [],[],[]

for r in resolutions:
  print(f"[{datetime.now()}]","Beanchmarking:", r)
  og.append(benchmark(og_set, r[0], r[1], x_min, x_max, y_min, y_max))
  c.append(benchmark(c_set, r[0], r[1], x_min, x_max, y_min, y_max))
  t.append(benchmark(t_set, r[0], r[1], x_min, x_max, y_min, y_max))
print(f"[{datetime.now()}]","done!")

plt.figure(figsize=(16, 9), dpi=300)
plt.plot(height, og, marker='o', label="Python")
plt.plot(height, c, marker='o', label="CPython")
plt.plot(height, t, marker='o', label="PyTorch")

plt.legend()
plt.yscale("log")
plt.xlabel("Resolution")
plt.ylabel("Execution time (s)")
plt.title("Mandelbrot Set: Python vs CPython vs PyTorch (100 max iterations)")
plt.grid(True)
plt.show()
```

```
[2026-02-26 15:48:50.313453] Beanchmarking: (256, 144)
[2026-02-26 15:48:52.827366] Beanchmarking: (426, 240)
[2026-02-26 15:48:59.057246] Beanchmarking: (640, 360)
[2026-02-26 15:49:11.908030] Beanchmarking: (853, 480)
```

```
[2026-02-26 15:49:34.756606] Beanchmarking: (1280, 720)
[2026-02-26 15:50:26.486596] Beanchmarking: (1920, 1080)
[2026-02-26 15:52:26.284227] Beanchmarking: (2560, 1440)
[2026-02-26 15:55:56.346638] done!
```



As can bee seen the pure Python implemnation losses overall which is to be expected. PyTorch was the fastest with CPython following behind, what's intresting is that at smaller gridsizes CPython preformed better than PyTorch. This is likely to the buss limitations and the time it takes to get the data to the GPU exceeeding the actual computation time.